Crop Recommendation Report

Lucknow, Uttar Pradesh • 2026

Top Recommended Crop	Sugarcane	Confidence: 29.6%
District	Lucknow	Uttar Pradesh
Assessment Year	2026	Crops Evaluated: 8

Recommendation Explanation

Sugarcane is the top recommendation for Lucknow in 2026. It has consistently delivered strong yields in this district, with a historical average of 68584 kg/ha and a previous-year yield of 67242 kg/ha. Its stable cultivation history makes it a dependable choice under current climate conditions.

Full Crop Ranking

Rank	Crop	Confidence (%)	Prev. Yield (kg/ha)	Hist. Mean (kg/ha)	Area Share (%)
1	**Sugarcane**	**29.6%**	**67242**	**68584**	**23.5%**
2	Wheat	11.6%	3376	3773	9.1%
3	Rice	11.1%	3113	2984	8.9%
4	Mustard	10.7%	1099	1291	6.6%
5	Lentil	10.4%	1087	935	8.0%
6	Maize	9.7%	2999	3010	10.1%
7	Chickpea	9.6%	1175	1155	16.9%
8	Groundnut	7.3%	1611	1585	16.8%

Key Influencing Factors

Factor	Influence Direction	Magnitude
Monsoon Season Precipitation	↑ Favourable	Medium
Growing Season Temperature	→ Neutral	Low
Previous Year Yield vs Historical Average	↓ Unfavourable	Medium
District Area Share	→ Neutral	Low
Irrigation Coverage Ratio	↑ Favourable	High

Climate Summary

Metric	Value
Annual Precipitation	990 mm
Monsoon Season Precipitation	848 mm
Average Maximum Temperature	32.0 °C

Average Minimum Temperature	18.1 °C
Irrigation Coverage Ratio	82.20%

This report was generated by the AgriCAST AI Crop Recommendation System using a LambdaMART Learning-to-Rank model trained on historical agricultural and climate data. Recommendations are decision-support tools and should be verified with local agronomists before operational use. AgriCAST v1.0